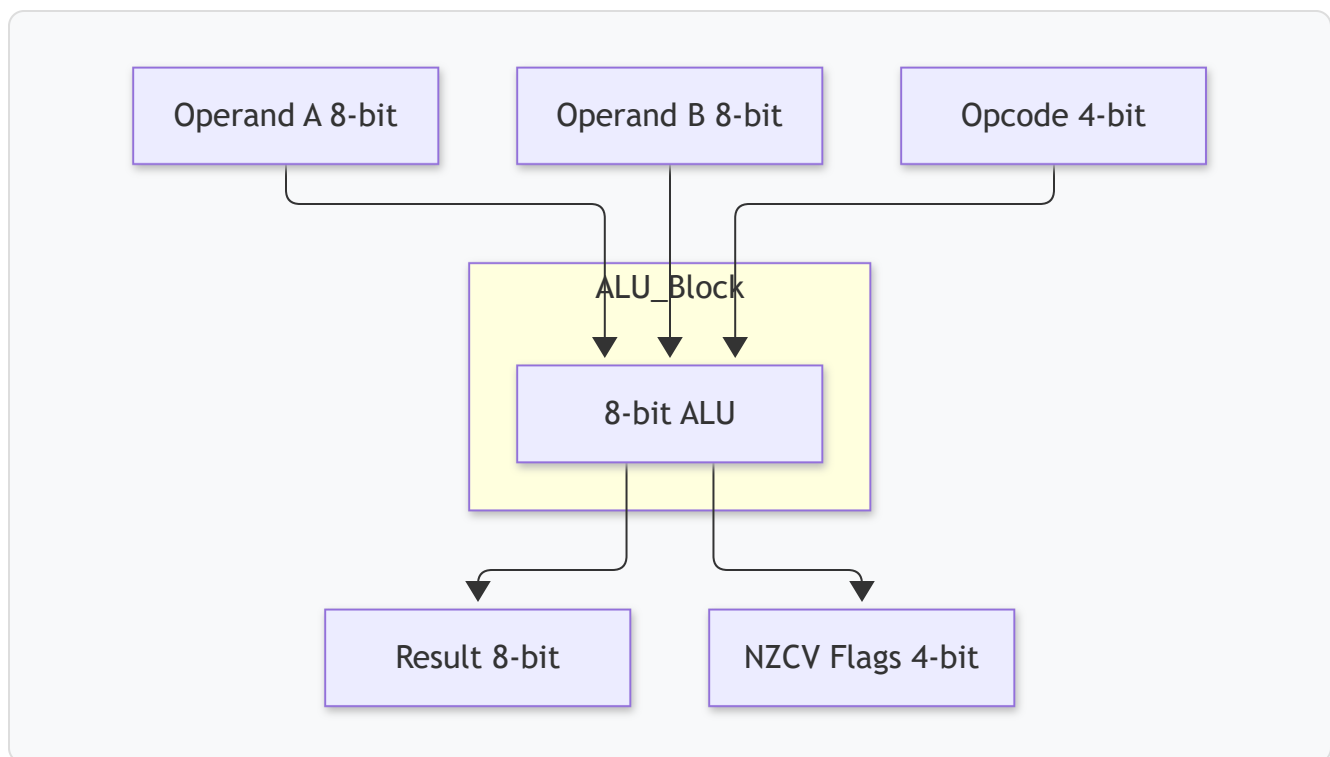


Experiment 1: Advanced 8-bit ALU with NZCV flags

1. Objective

To design, simulate, and implement an Advanced 8-bit Arithmetic Logic Unit (ALU) capable of performing various arithmetic and logical operations while generating Negative (N), Zero (Z), Carry (C), and Overflow (V) flags.

2. Block Diagram

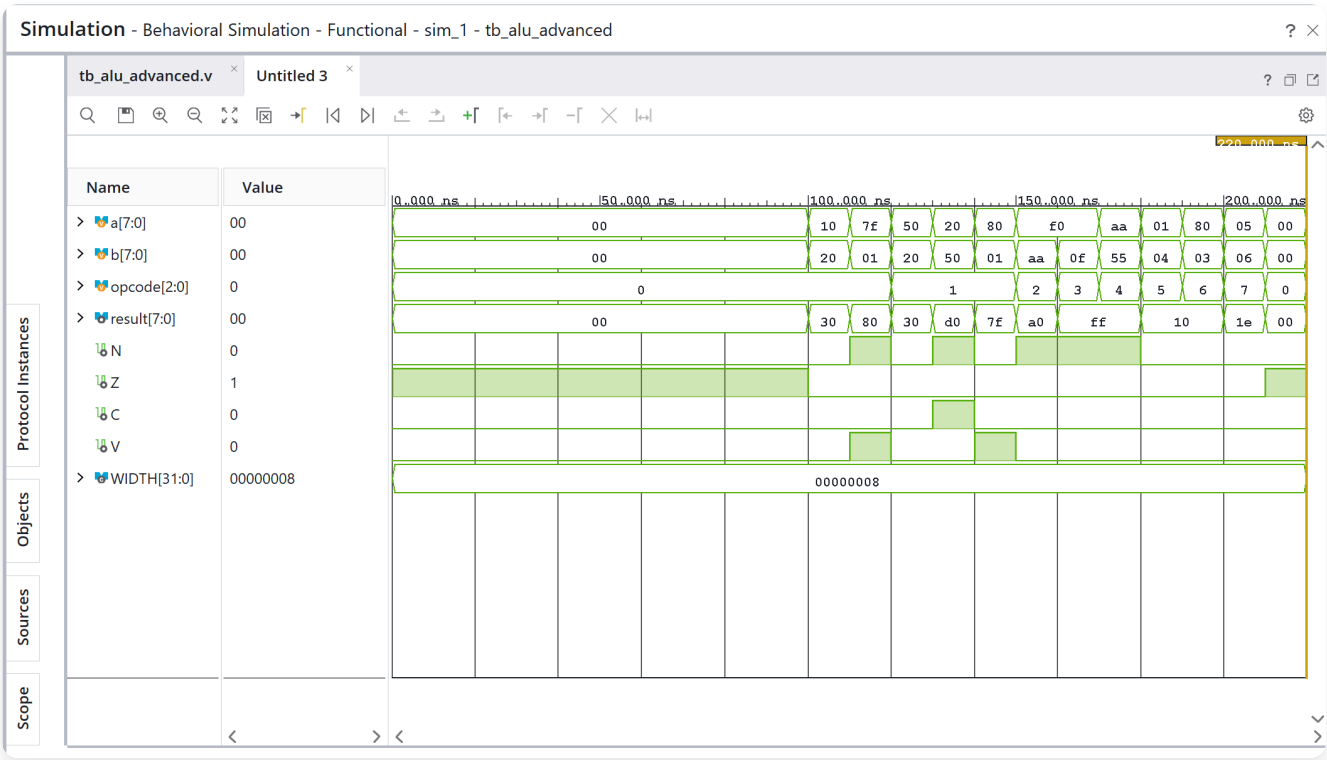


3. RTL Design

The design consists of an 8-bit ALU that supports operations such as ADD, SUB, AND, OR, XOR, NOT, and shifts. - **8-bit ALU**: The core logic performs the selected operation based on a 4-bit Opcode. - **NZCV Flags**: - **N (Negative)**: Set if the MSB of the result is 1. - **Z (Zero)**: Set if

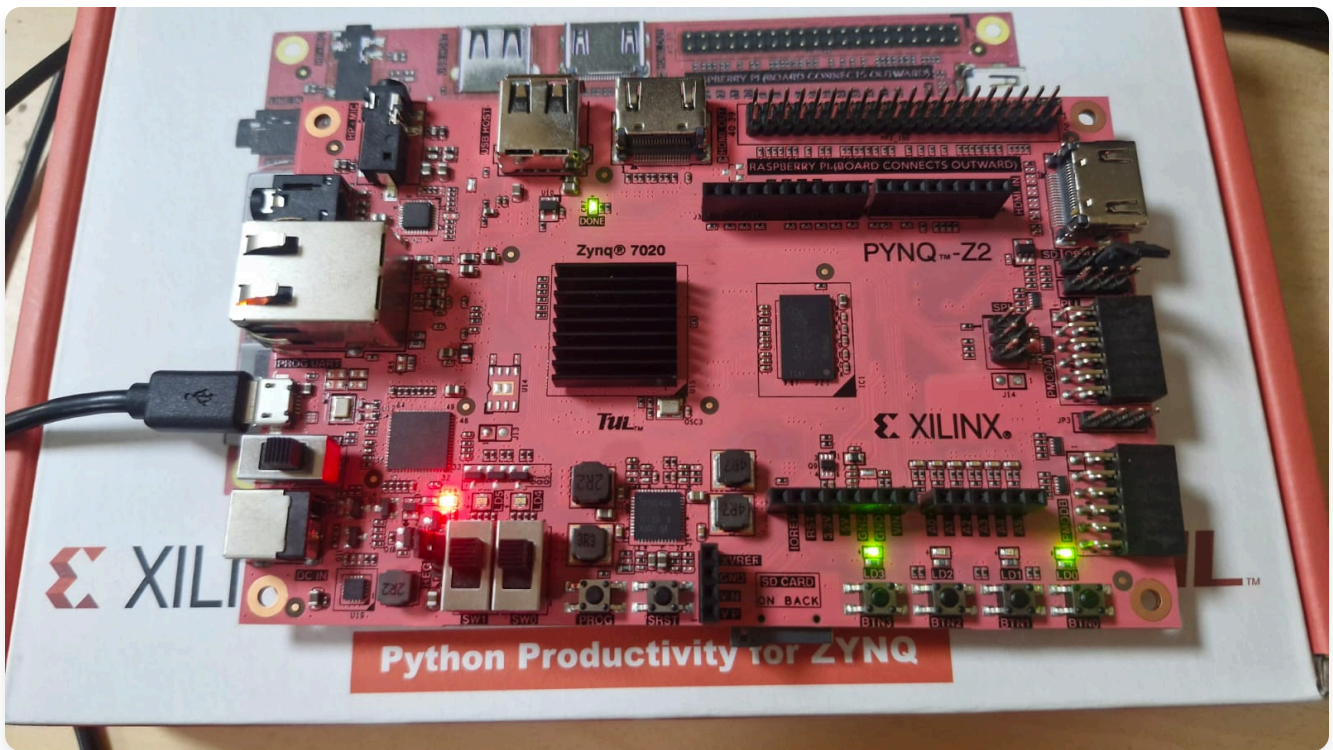
the result is exactly zero. - **C (Carry)**: Set if an arithmetic operation generates a carry out from the MSB. - **V (Overflow)**: Set if an arithmetic operation results in a signed overflow. - **PYNQ-Z2 Auto-Test Wrapper**: A wrapper module designed for the PYNQ-Z2 board to facilitate automated testing by mapping the ALU I/O to the board's switches, buttons, and LEDs for hardware validation.

4. Simulation Results



- Screenshot 1: Addition and Subtraction with N and C flags. - Screenshot 2: Logical operations with N and Z flags. - Screenshot 3: Overflow condition during signed arithmetic.

5. Hardware Implementation



- Photo 1: PYNQ-Z2 board showing ALU inputs via switches. - Photo 2: PYNQ-Z2 board showing output result and flags on LEDs.

6. Applications

- Central Processing Unit (CPU) ALUs
- Digital Signal Processing (DSP) applications
- Embedded microcontrollers

7. Conclusion

An 8-bit Advanced ALU with accurate NZCV flag generation was successfully designed and tested. The implementation on the PYNQ-Z2 board verifies the hardware functionality, making it a reliable building block for more complex computational architectures.